

Physical Chemistry (I)

Lecture 15

Phase Transitions



In the Last Class

- Applications of statistical mechanics
 - Perfect gas
 - Equipartition theorem
- Phases
 - Phase space
 - Calculation of partition function

Perfect Gas

- No interactions (no potential energy)
- Total energy = kinetic energy ($p^2/2m$)
- Bound in a box of volume V
 - Range of $x = (0, a)$
 - Range of $y = (0, b)$
 - Range of $z = (0, c)$
 - $abc = V$
- Range of momentum = $(-\infty, \infty)$

Last class

Q of Perfect Gas

$$q = V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2}$$

For indistinguishable gas particles,

$$Q = \frac{1}{N!} q^N = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Last class

Thermodynamics

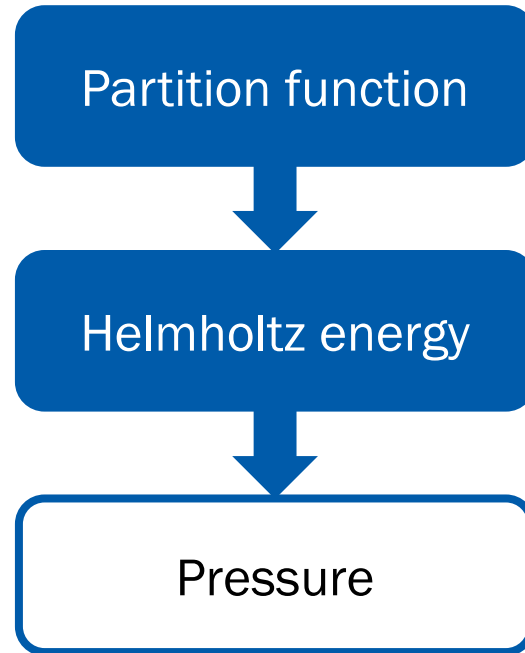
$$Q = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

$$A = -Nk_B T \ln V + k_B T \ln N! - \frac{3}{2} Nk_B T \ln \left(\frac{2\pi m k_B T}{h^2} \right)$$

$$U = \frac{3}{2} Nk_B T$$

$$p = \frac{Nk_B T}{V}$$

Pressure of Real Gas



$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

Last class

Quadratic Contributions

$$Q = \frac{1}{N! h^{3N}} \int \cdots \int e^{-E_{\text{tot}}/k_{\text{B}}T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

Assume that

$$E_{\text{tot}} = \sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2$$
$$(A_i, B_j > 0)$$

and that the ranges of both positions and momenta are $(-\infty, \infty)$

Equipartition Theorem

$$Q = \frac{\pi^{(f_r+f_p)/2}}{N! h^{3N}} \left(\frac{1}{\beta}\right)^{(f_r+f_p)/2} \prod_{i=1}^{f_r} \left(\frac{1}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{1}{B_j}\right)^{1/2}$$

$$U = (f_r + f_p) \frac{k_B T}{2}$$

“The average value of each **quadratic contribution** to the internal energy is $(1/2)k_B T$.”

Last class

Phase

A form of matter that is **uniform** throughout
in **chemical composition** and **physical state**

Phase Transition

Spontaneous conversion
of one phase into another phase

- occurs at transition temperature T_{trs} (for given p)
- At T_{trs} , the two phases are at equilibrium

Fundamental Equation

$$dU = TdS - pdV + \mu dn$$

- Mechanistic equilibrium: $p_1 = p_2$
 - Conjugate extensive variable: dV
- Thermal equilibrium: $T_1 = T_2$
 - Conjugate extensive variable: $dS = \delta q/T$
- Diffusive equilibrium: $\mu_1 = \mu_2$
 - Conjugate extensive variable: dn



Last class

Meaning of μ

$$\mu = \left(\frac{\partial G}{\partial n} \right)_{T,p} = \frac{G}{n} = \mu(T, p)$$

(molar Gibbs energy)

Thermodynamic Potentials

$$G = \mu n$$

$$H = G + TS = \mu n + TS$$

$$A = G - pV = \mu n - pV$$

$$U = G + TS - pV = \mu n + TS - pV$$

- Gibbs-Duhem equation

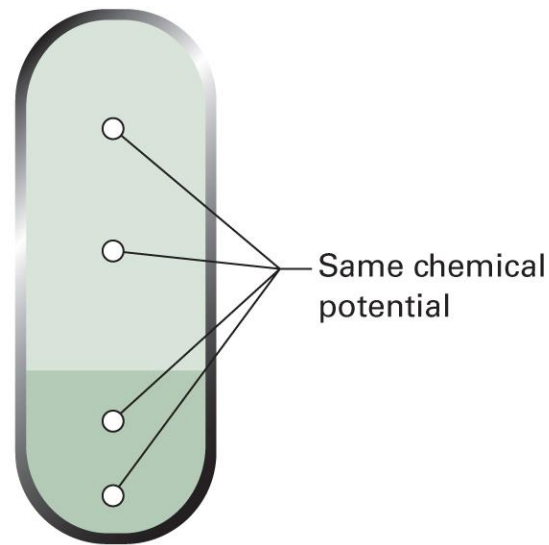
$$SdT - Vdp + nd\mu = 0$$

$$d\mu = -S_m dT + V_m dp$$

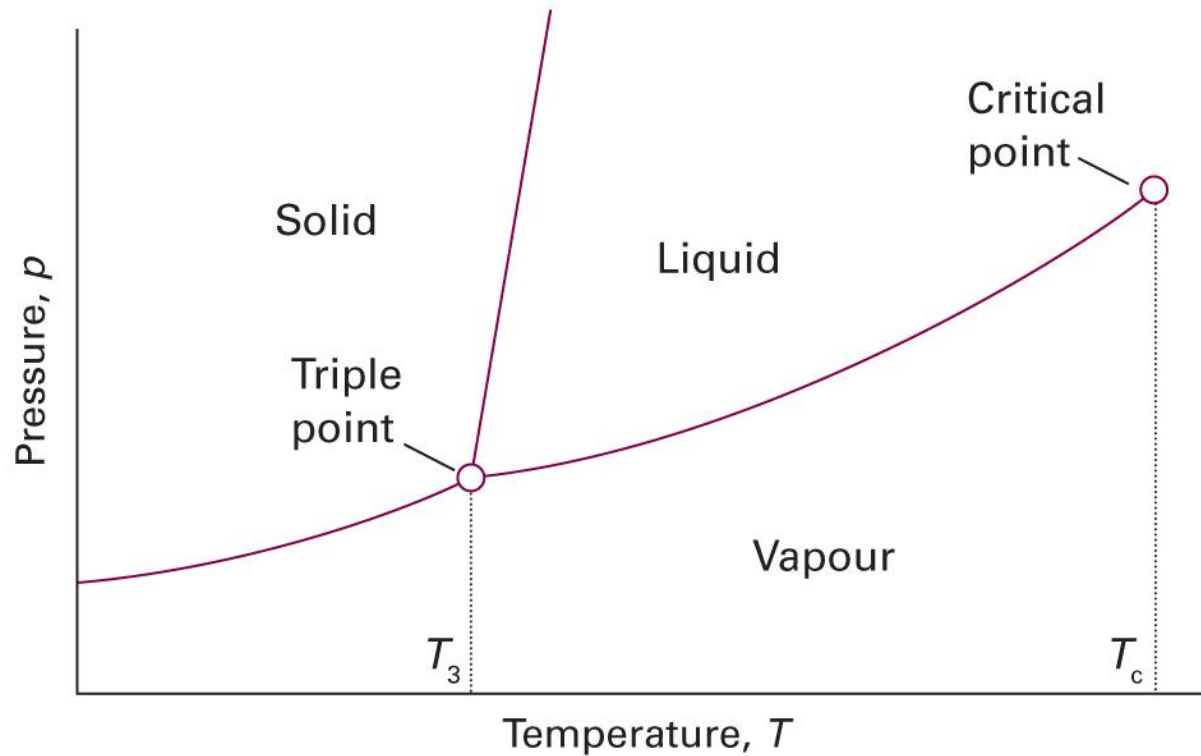
Last class

Criteria for Phase Stability

At equilibrium, the **chemical potential** of a substance is the **same** in and throughout every phase in the system



Typical Phase Diagram



Phase Rule

“Can we have a quadruple point?”

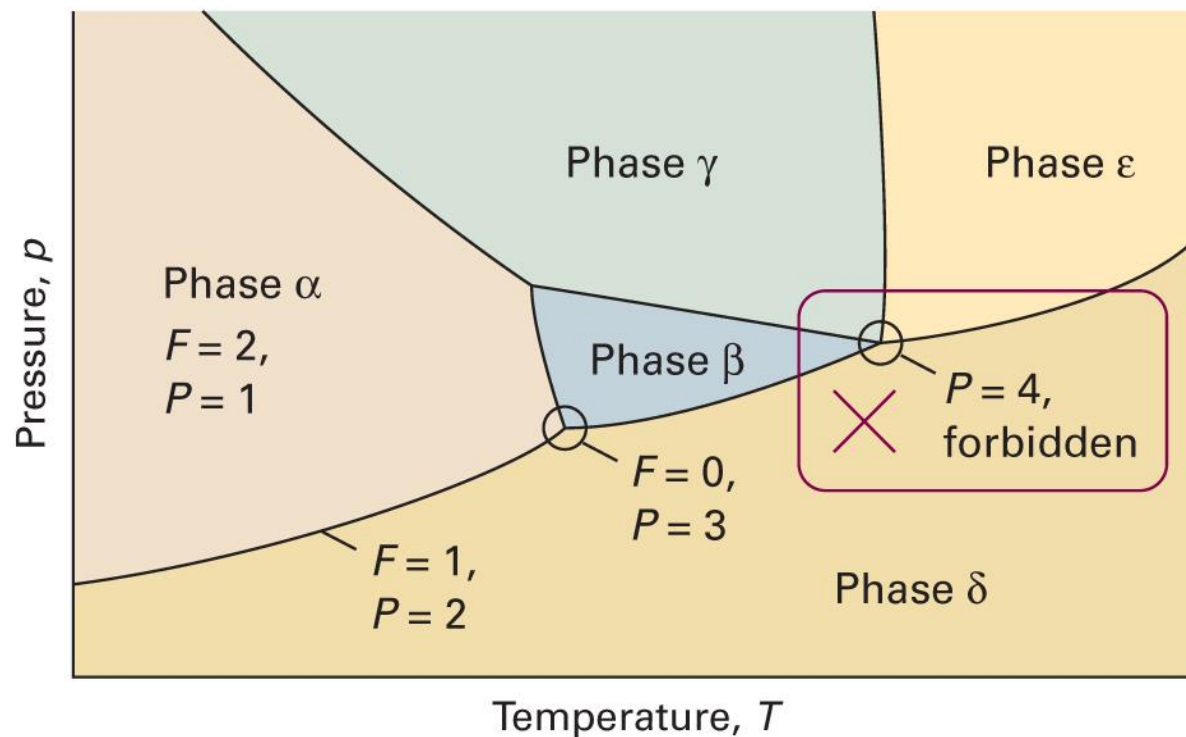
- **Phase rule:** a general relation between
 - Number of degrees of freedom F
 - Number of components C
 - Number of phases at equilibrium P
- Use the **criteria for phase stability** (same μ)
 - Recall that $\mu = g(T, p)$

Single Component ($C = 1$)

- $P = 1$: both T and p can be independently varied
 $\therefore F = 2$ (bivariant)
- $P = 2$: $\mu_1(T, p) = \mu_2(T, p)$
 $\therefore F = 1$ (univariant)
- $P = 3$: $\mu_1(T, p) = \mu_2(T, p)$ and $\mu_2(T, p) = \mu_3(T, p)$
 $\therefore F = 0$ (invariant)
- $P = 4$: two variables and three restrictions $\rightarrow$ impossible

In summary, $F = 3 - P$

Single Component ($C = 1$)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.7)

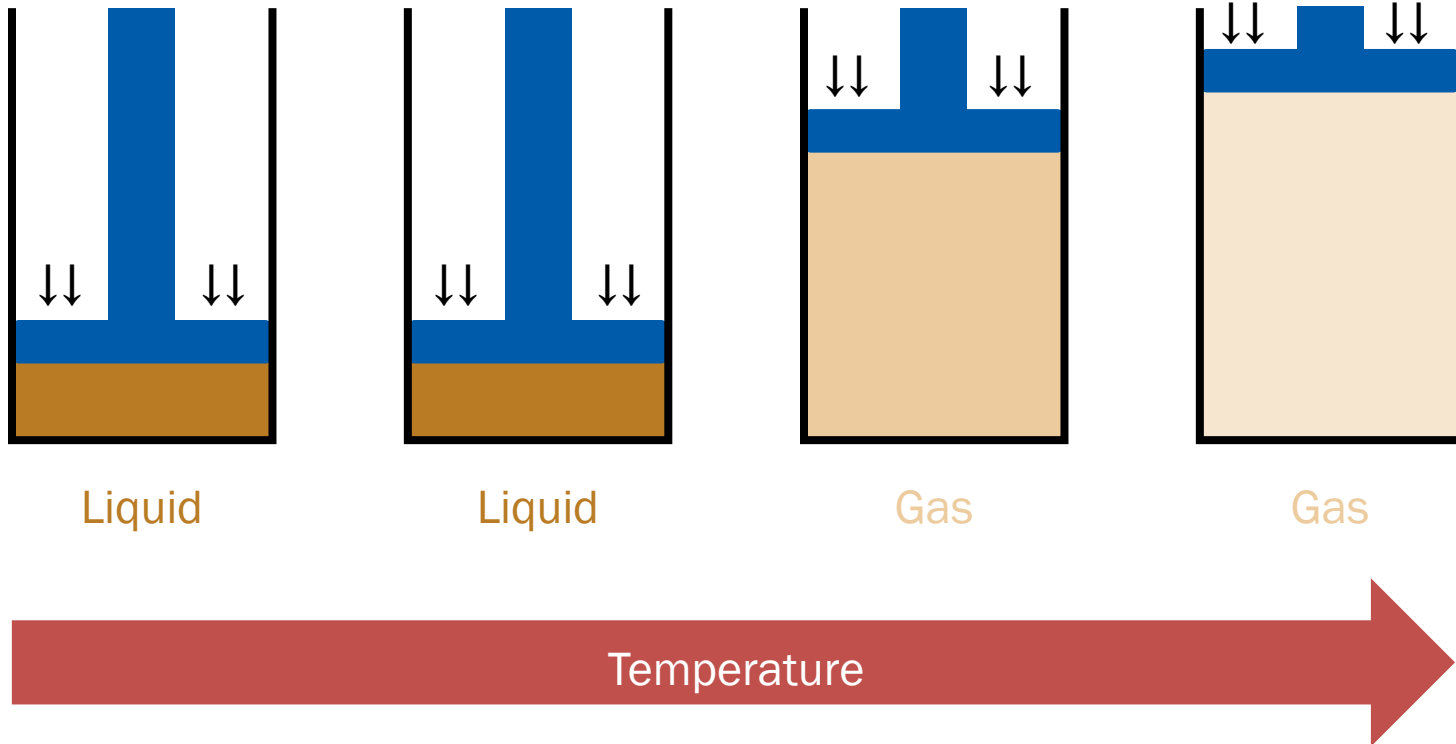
Multiple Components ($C > 1$)

- The total number of independent intensive variables
 - T and p : 2
 - $(C - 1)$ mole fractions (-1 due to normalization) for each phase: $P(C - 1)$
 - Total number = $P(C - 1) + 2$
- The total number of equations (restrictions)
 - $\mu_J(\text{phase 1}; p, T) = \mu_J(\text{phase 2}; p, T) = \dots$ for component J
 - Total number = $C(P - 1)$
- Therefore, $F = P(C - 1) + 2 - C(P - 1) = C - P + 2$

Not in Atkins

Liquid-Gas Transition 1

(Isobaric change)

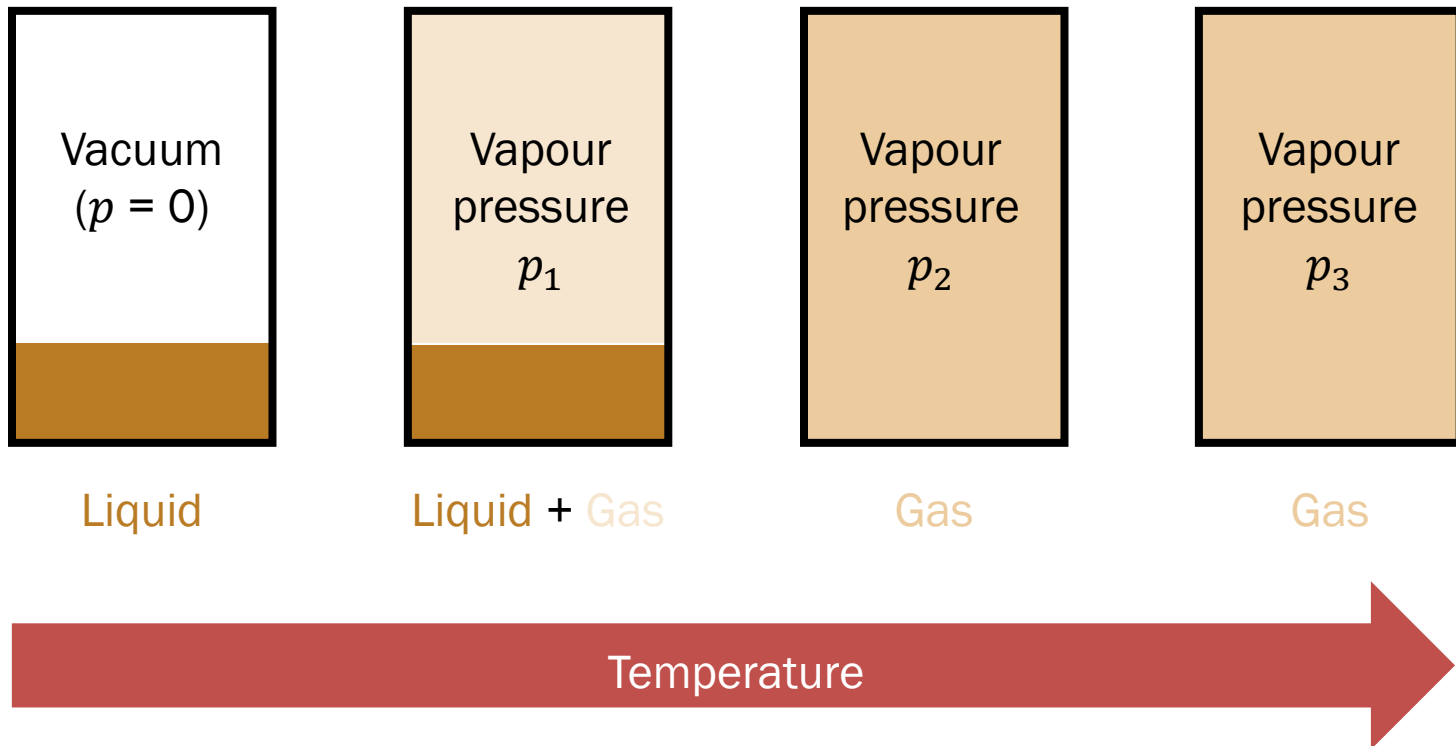


Not in Atkins

Liquid-Gas Transition 2

(Isochoric change)

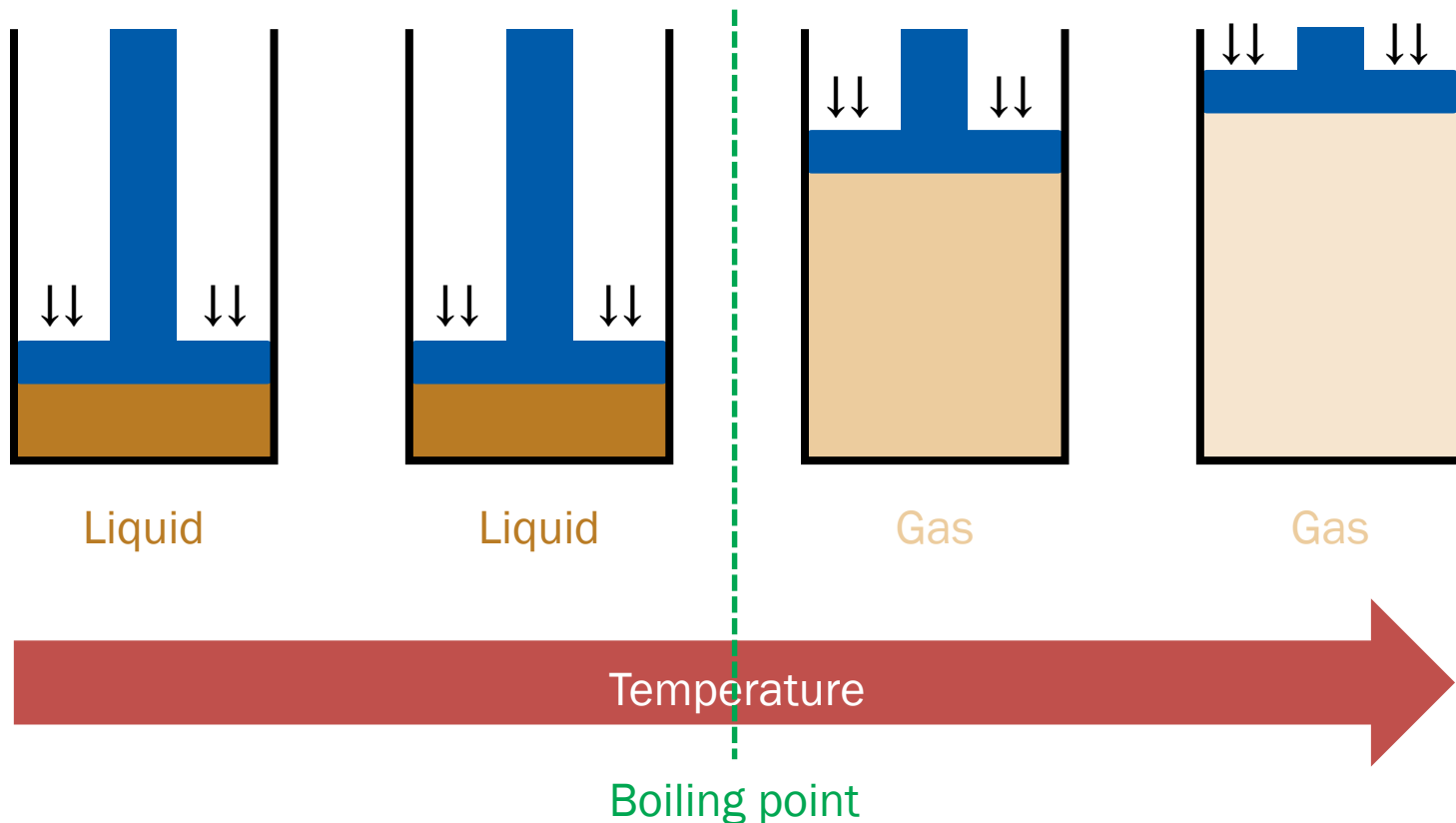
$(p_1 < p_2 < p_3)$



Not in Atkins

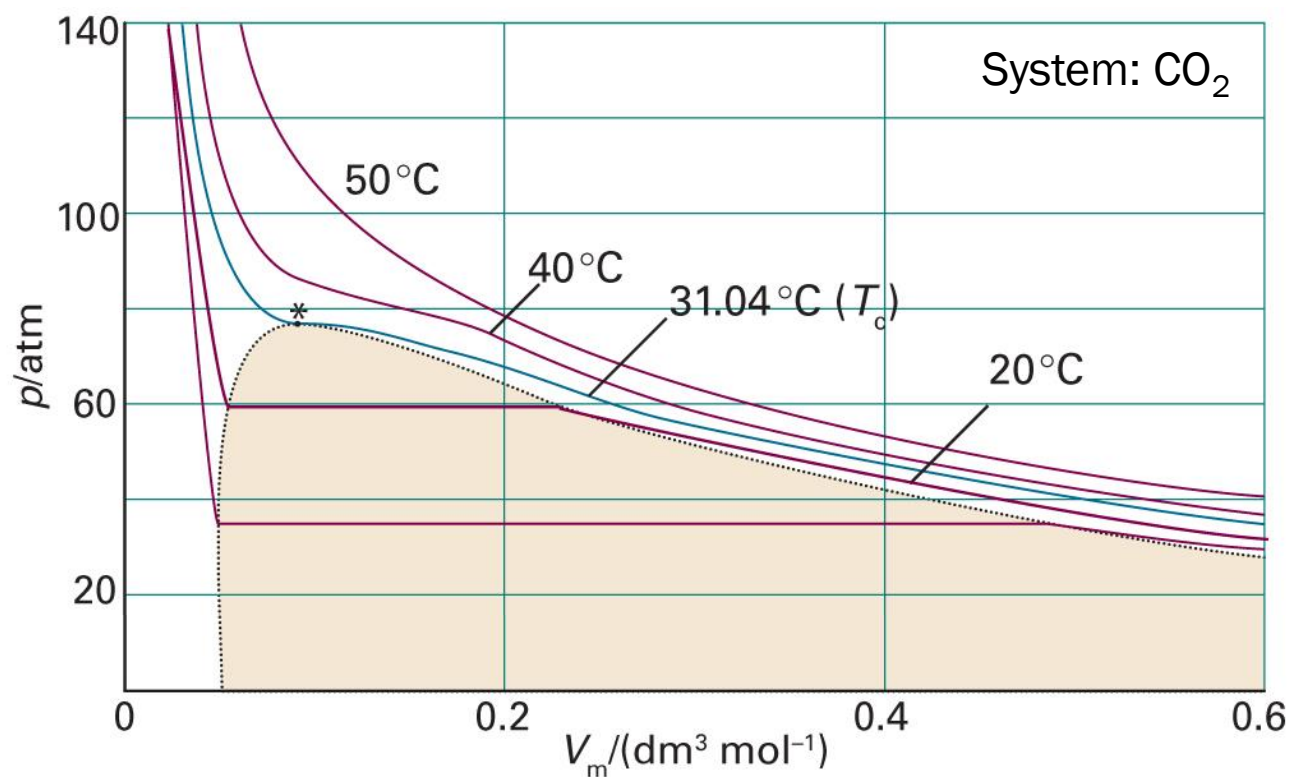
Vapor Pressure and Boiling

(Isobaric change)



Liquid-Gas Transition 3

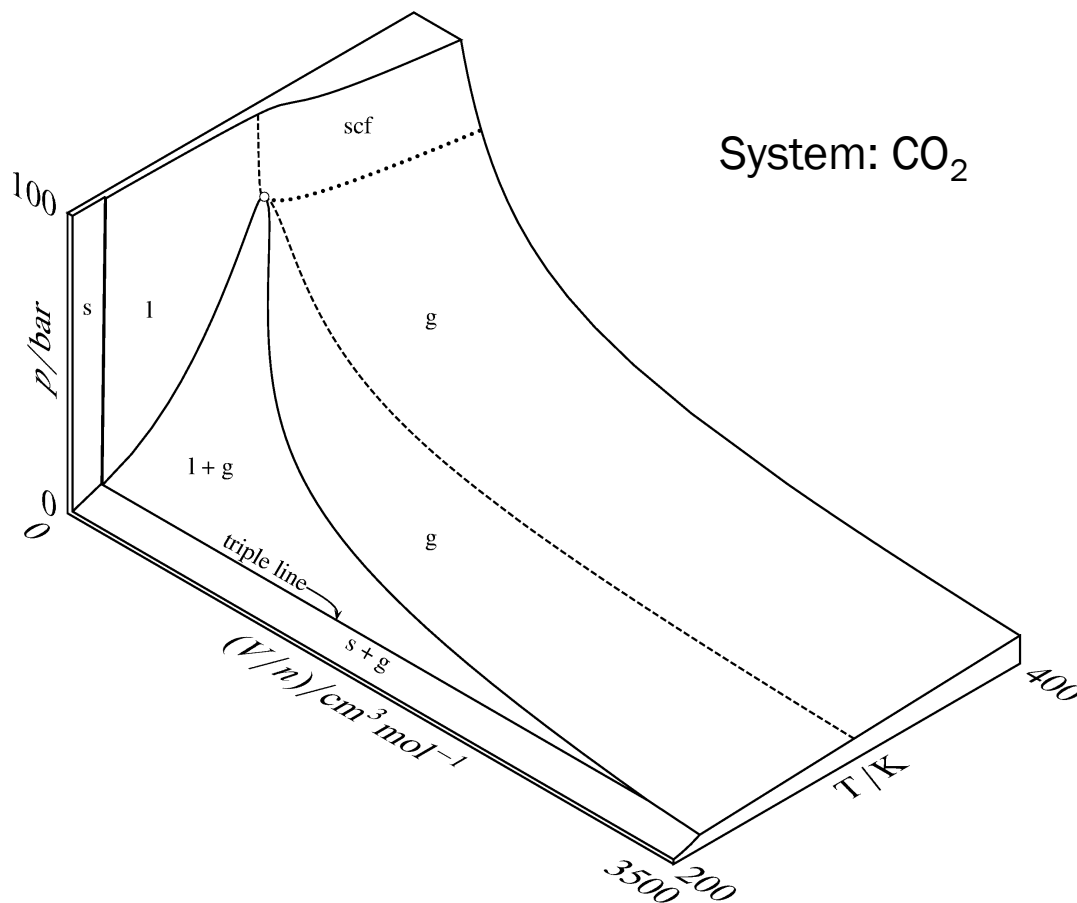
(Isothermal change)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 1C.2)

Not in Atkins

3-Dimensional Surface



(Image: <https://chem.libretexts.org/@go/page/20608>)

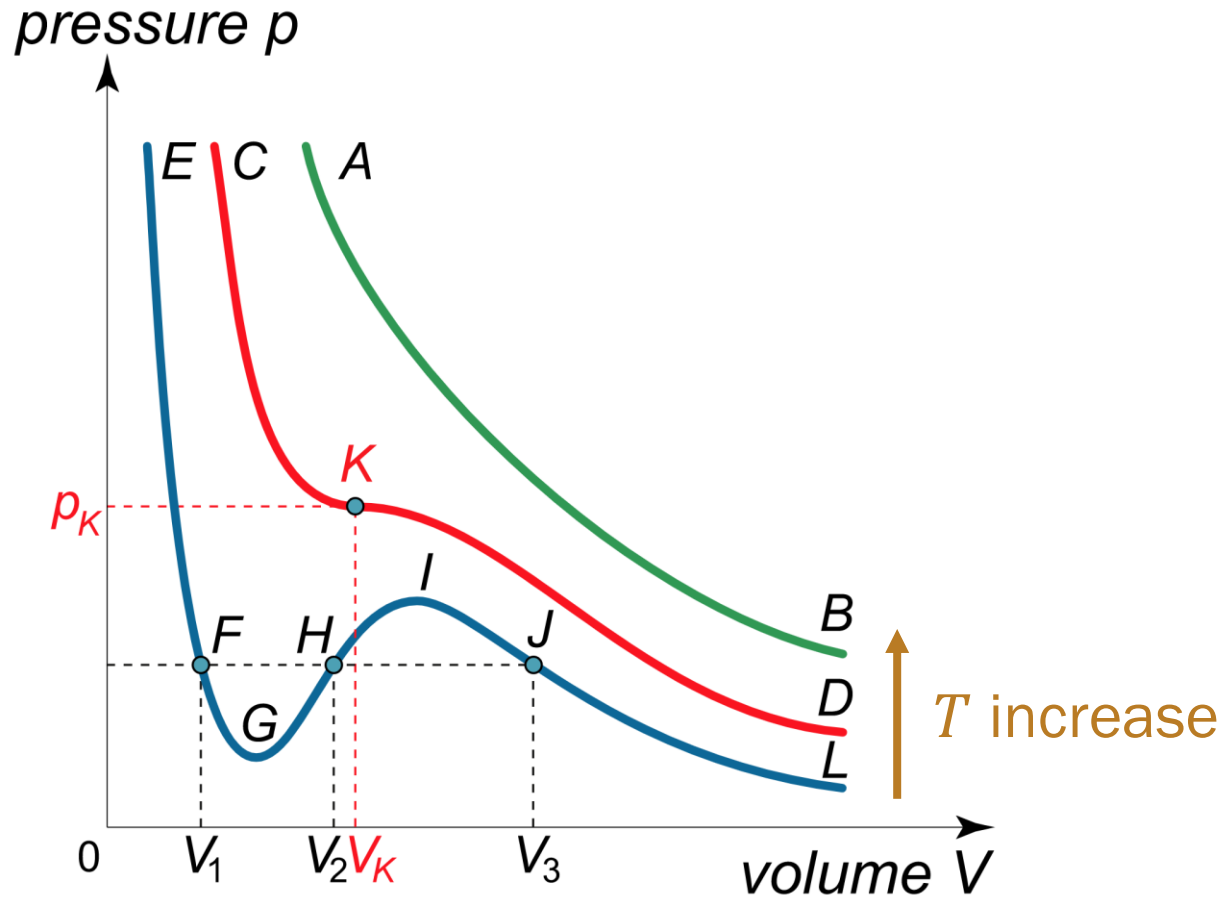
Good Old Friend

“Can we describe the liquid-gas transition using a model system?”

The van der Waals fluid!

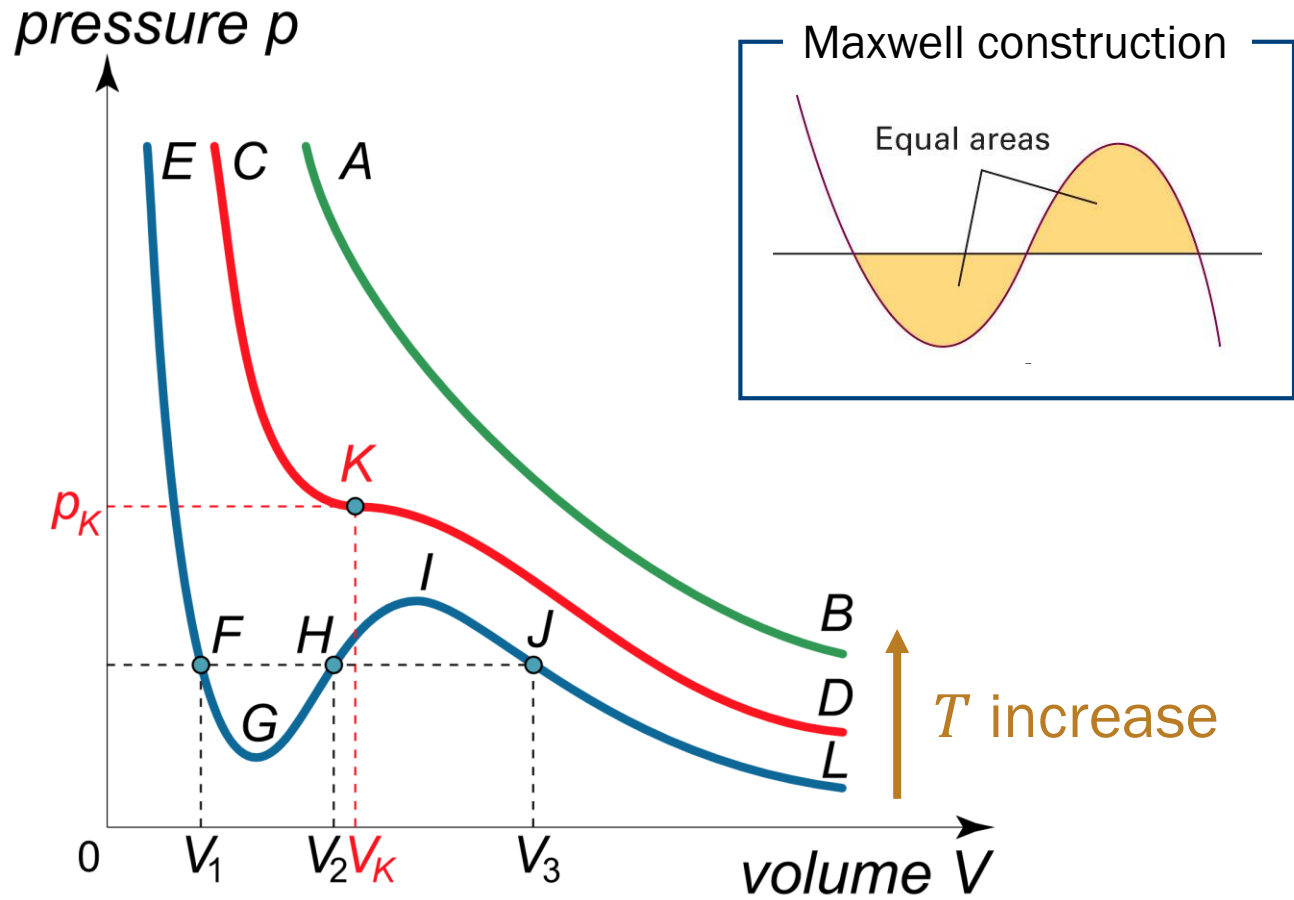
$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

van der Waals Isotherms



(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

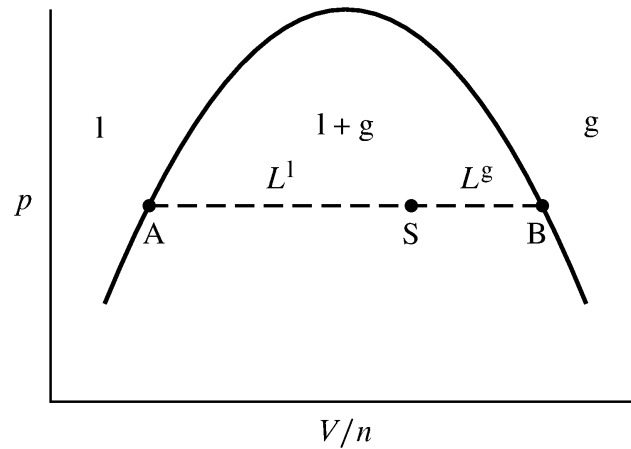
Curve E-L: Liquid-Gas Transition



(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

Not in Atkins

What Happens on Tie Line?

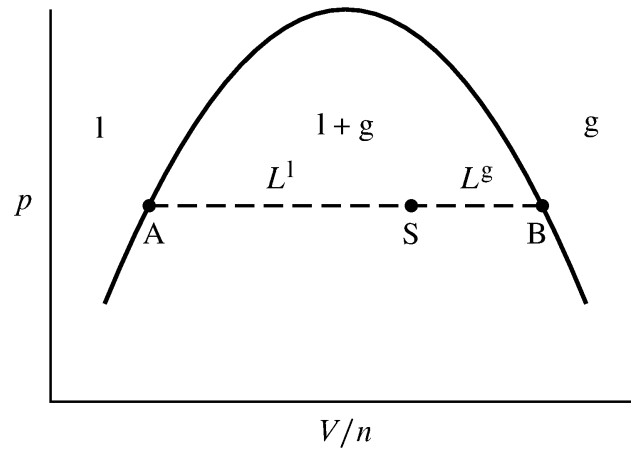


- Total amount at point S: $n = n_l + n_g$
- Total volume at point S: $V = V_{m,l}n_l + V_{m,g}n_g$

$$\frac{V}{n} = \frac{V_{m,l}n_l + V_{m,g}n_g}{n_l + n_g}$$

Not in Atkins

What Happens on Tie Line?



$$n_l \left(V_{m,l} - \frac{V}{n} \right) = n_g \left(\frac{V}{n} - V_{m,g} \right)$$

$$n_l L^l = n_g L^g$$

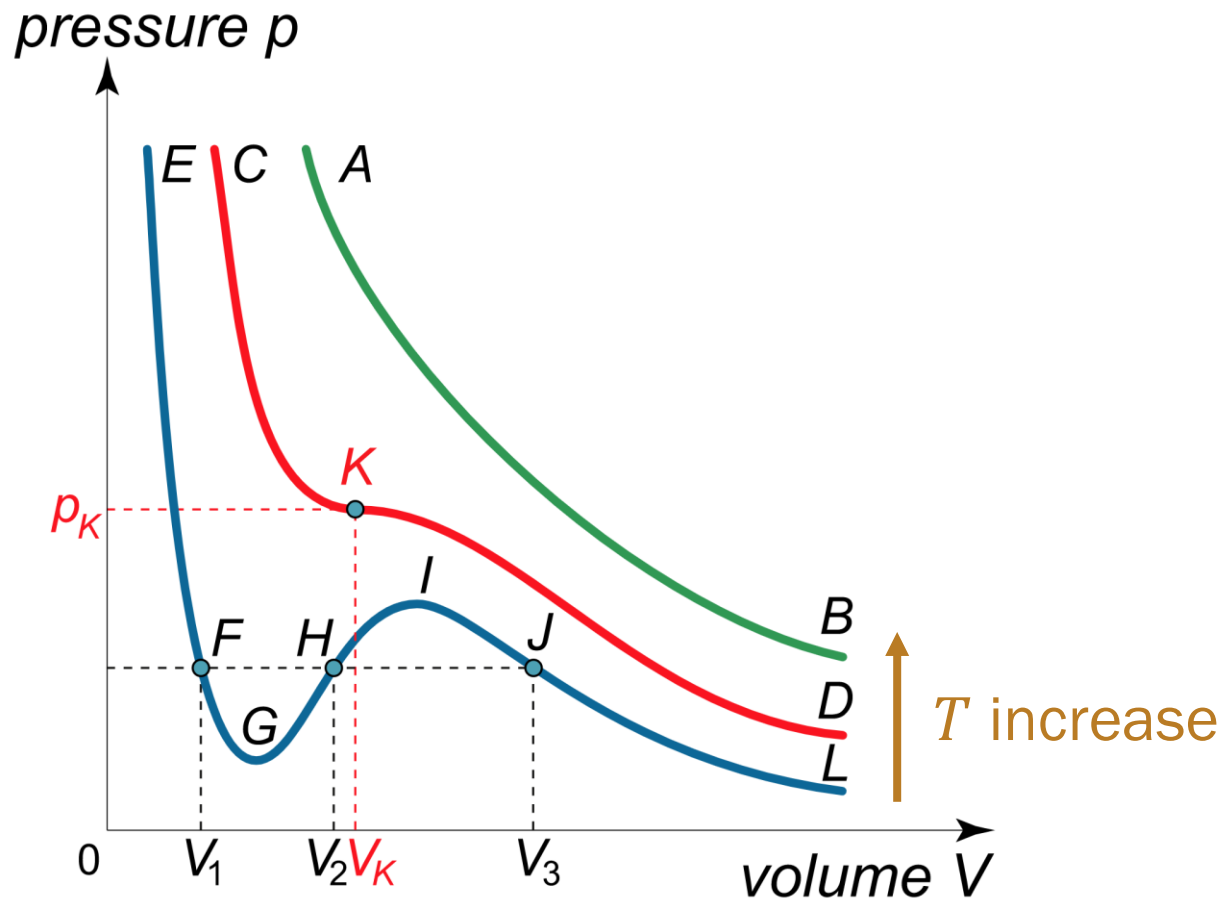
$$n_l : n_g = L^g : L^l \text{ (lever rule)}$$



(Image: <https://chem.libretexts.org/@go/page/20608>)

Not in Atkins

F-G and I-J: What Happens?



(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

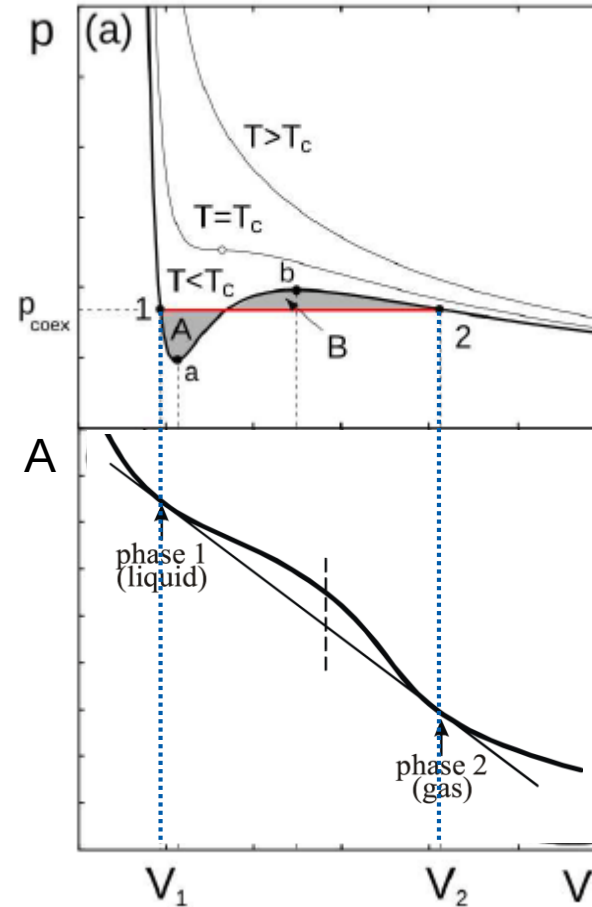
Not in Atkins

Metastable Phases

$$p = - \left(\frac{\partial A}{\partial V} \right)_{T,n}$$

At constant T and n ,

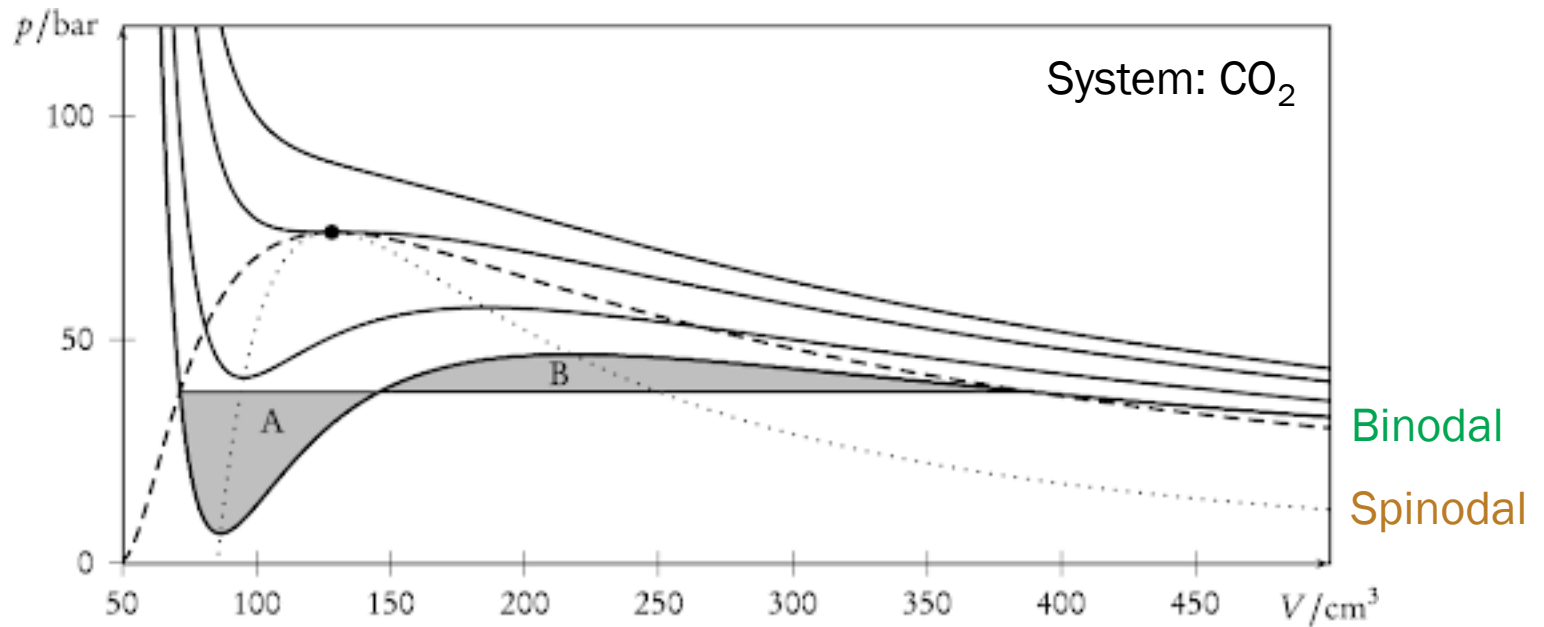
$$A = - \int p \, dV$$



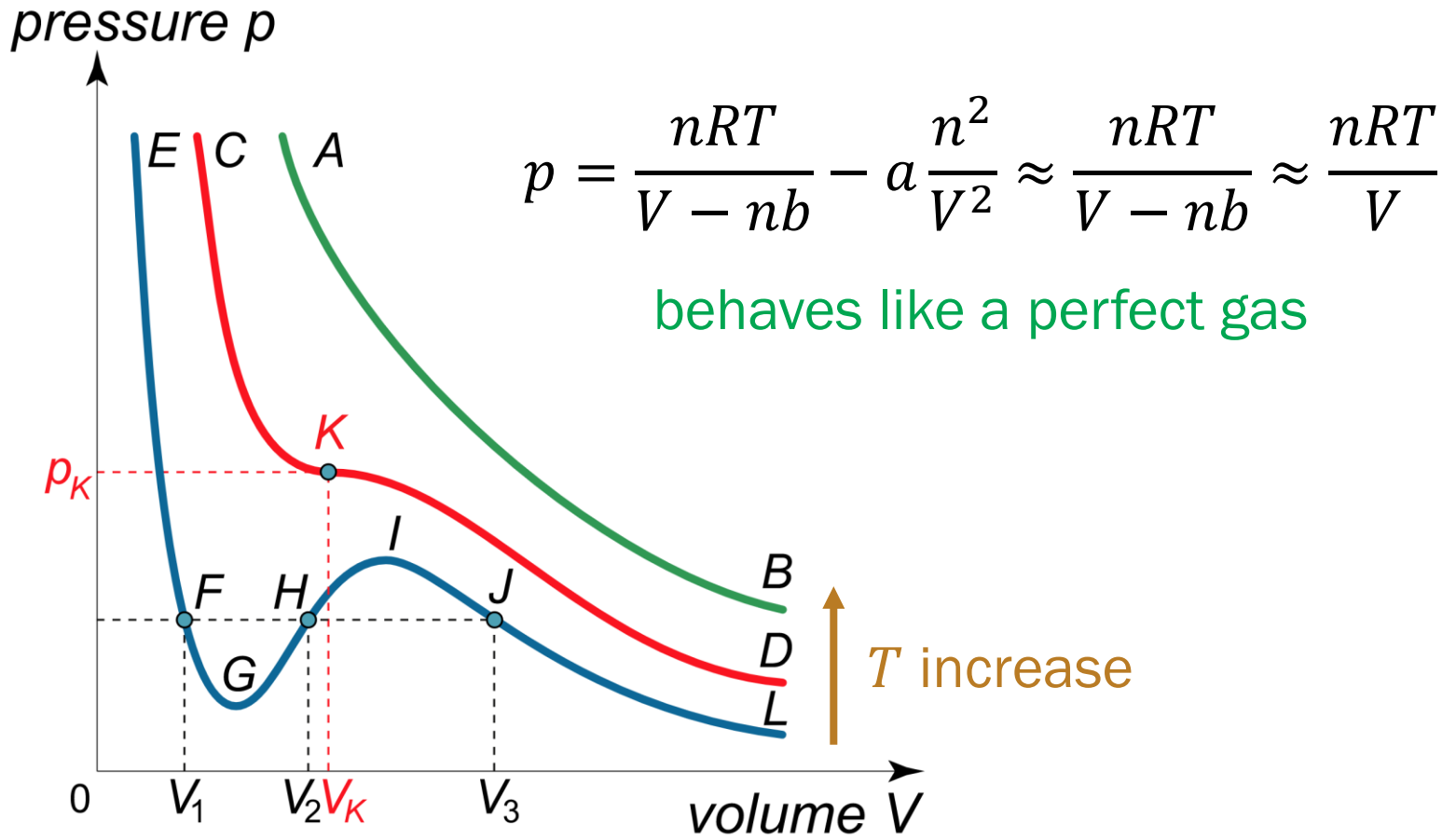
(Image: <https://www.physicsforums.com/threads/liquid-gas-phase-transition-metastable-mixed-states.935905>) 31

Not in Atkins

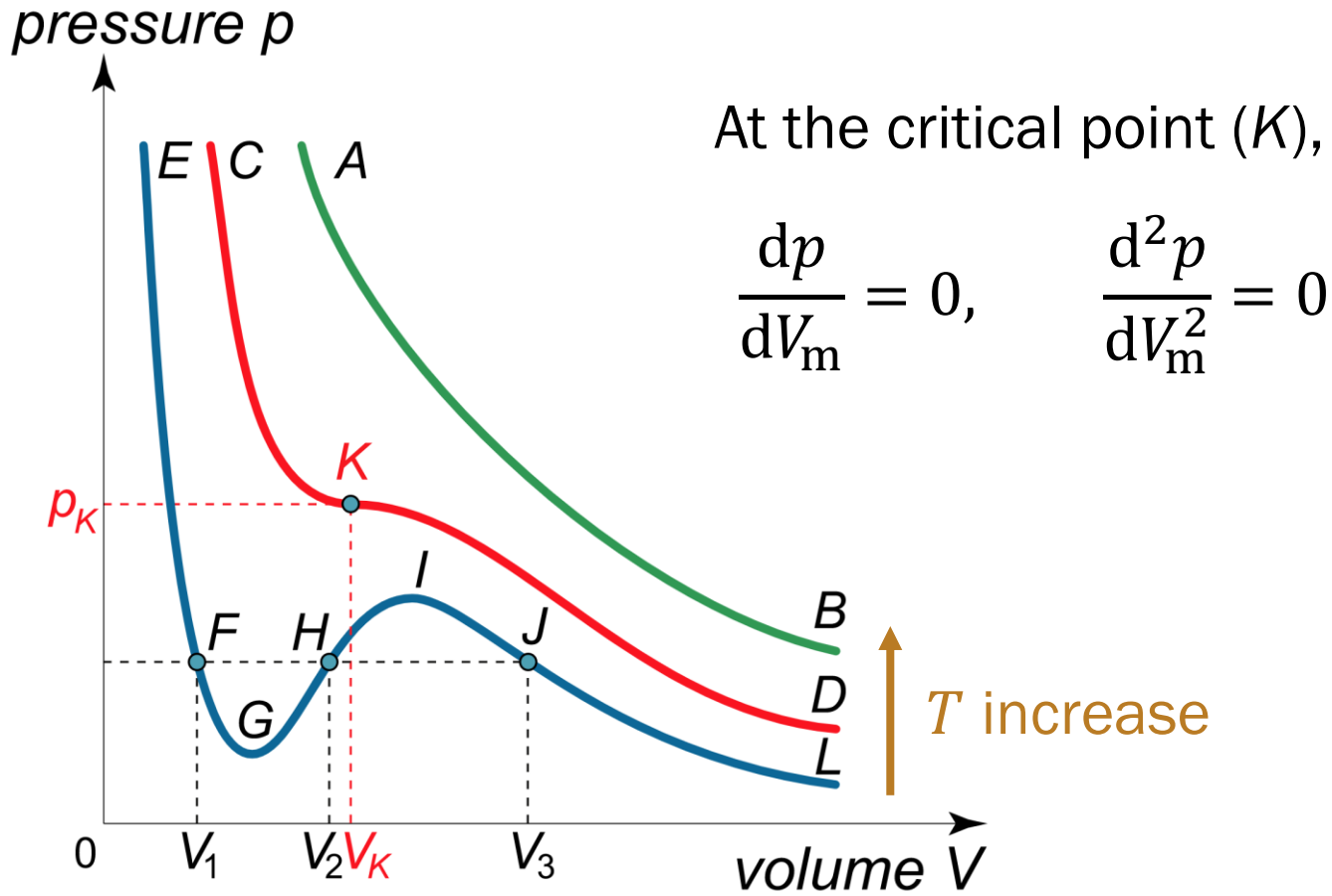
Binodal and Spinodal Lines



Curve A-B: Supercritical Fluid



Curve C-D: Critical Point



(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

Critical Constants

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2} = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$\frac{dp}{dV_m} = -\frac{RT}{(V_m - b)^2} + \frac{2a}{V_m^3} = 0$$

$$\frac{d^2p}{dV_m^2} = \frac{2RT}{(V_m - b)^3} - \frac{6a}{V_m^4} = 0$$

Solving these equations, one can obtain

$$V_c = 3b, \quad p_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27bR}$$

Critical Compression Factor

$$V_c = 3b, \quad p_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27bR}$$

The critical compression factor is defined as

$$Z_c = \frac{p_c V_c}{RT_c} = \frac{3}{8} = 0.375$$

Experimental values of Z_c :

Ar – 0.292, CO₂ – 0.274, He – 0.305, O₂ – 0.308

Reduced Variables

$$p_r = p/p_c$$

$$V_r = V_m/V_c$$

$$T_r = T/T_c$$

Variables without the substance-specific features

Rewriting vdW Equation

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

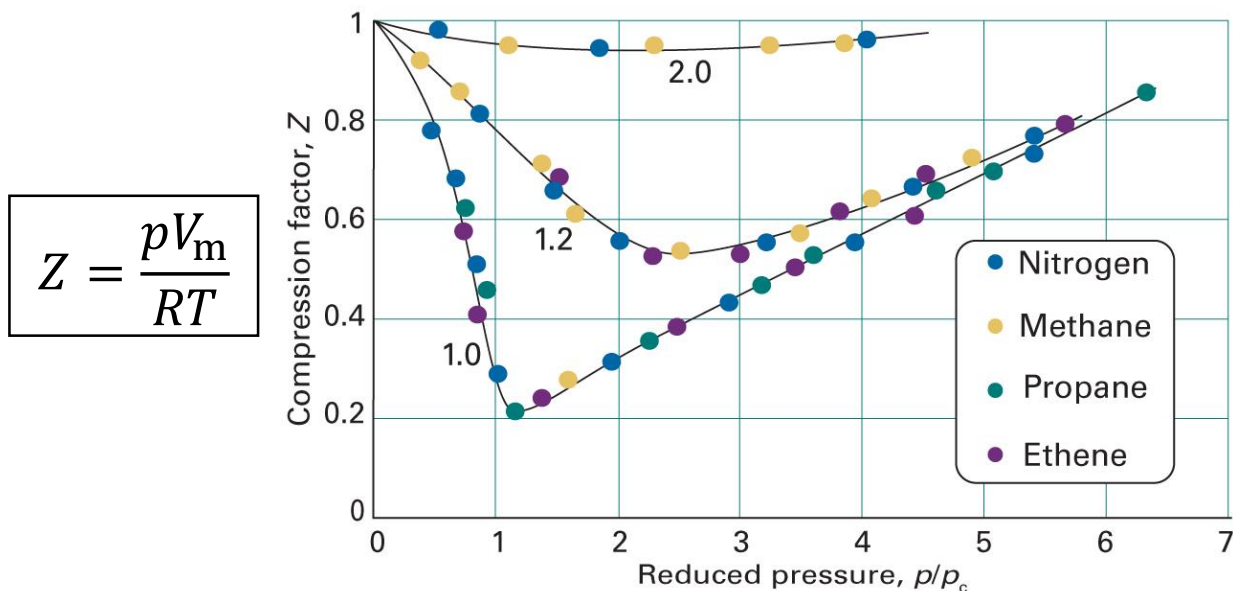
$$p_r p_c = \frac{RT_r T_c}{V_r V_c - b} - \frac{a}{V_r^2 V_c^2}$$

$$\frac{ap_r}{27b^2} = \frac{8aT_r}{27b(3bV_r - b)} - \frac{a}{9b^2 V_r^2}$$

$$p_r = \frac{8T_r}{3V_r - 1} - \frac{3}{V_r^2}$$

Principle of Corresponding States

Gases at the same V_r and T_r exert the same p_r



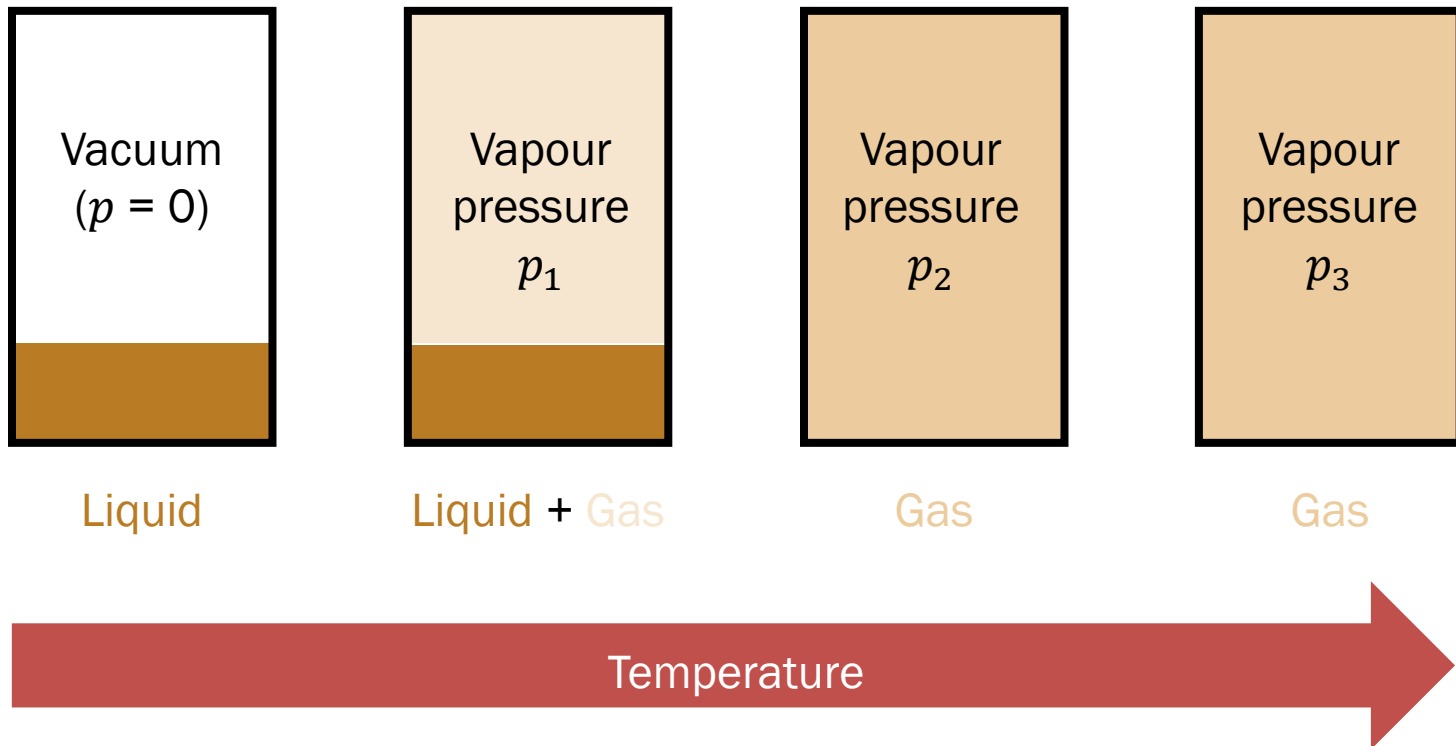
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 1C.8)

Not in Atkins

Coexistence Region

(Isochoric change)

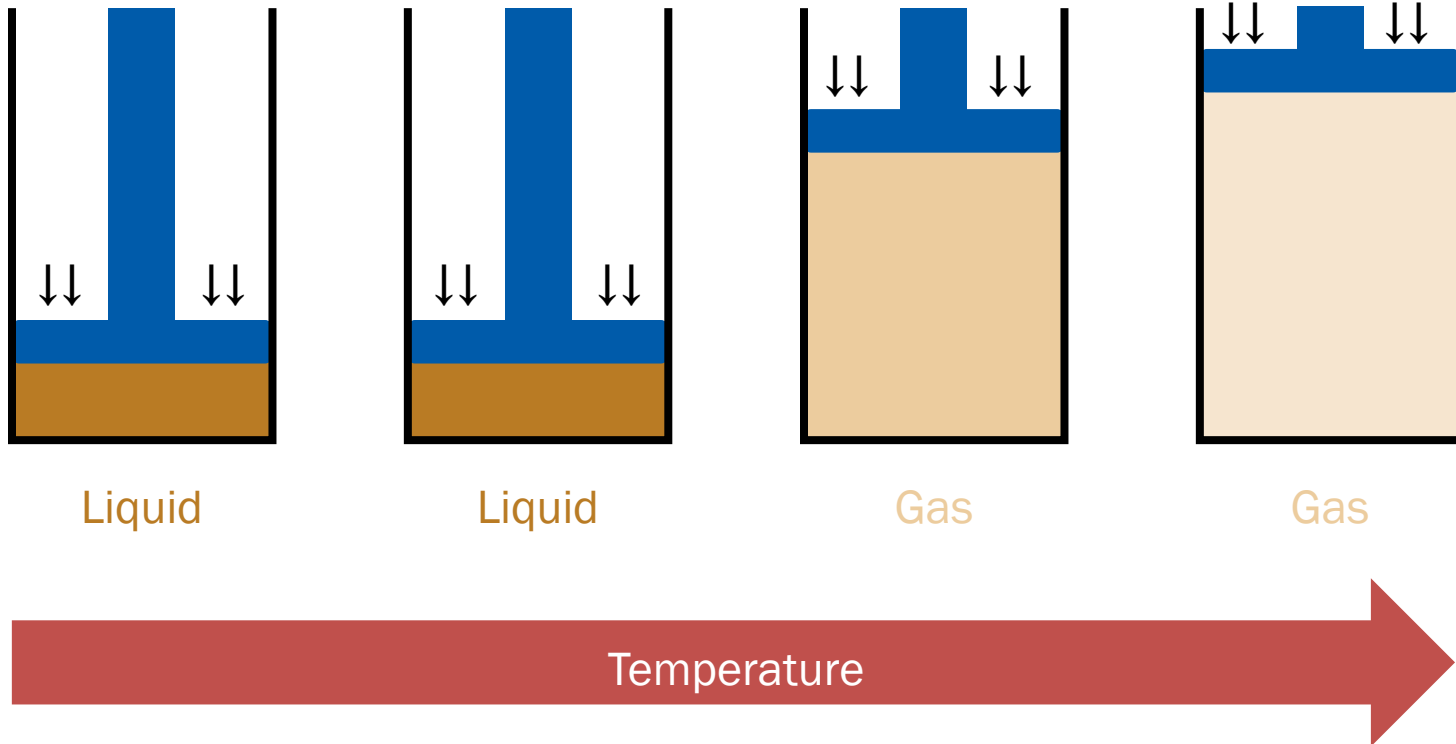
$(p_1 < p_2 < p_3)$



Not in Atkins

No Coexistence Region

(Isobaric change)



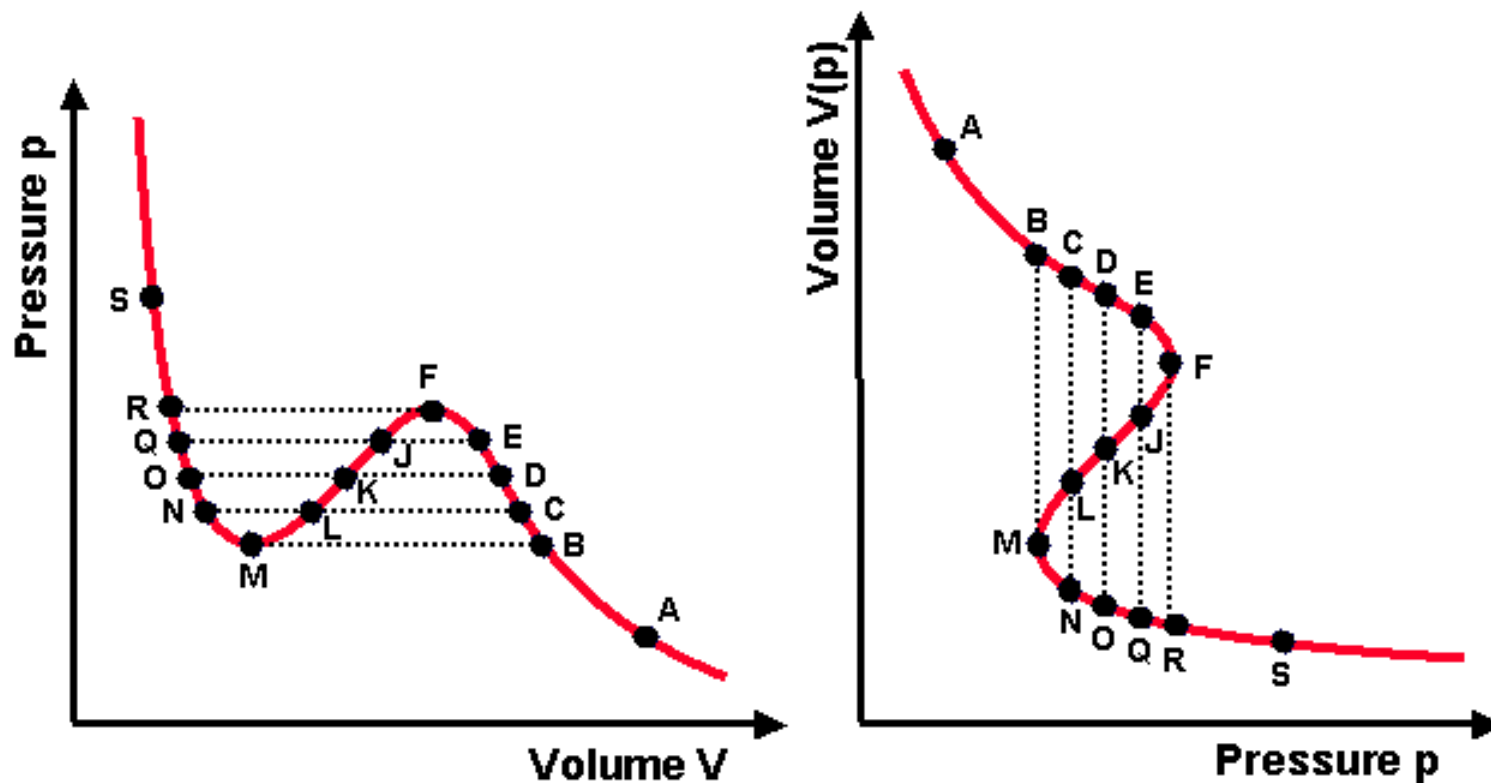
Not in Atkins

Transitions of vdW Fluid

Fixed	Independent	Dependent	Coexistence?
V	T	p	Yes
p	T	V	No
T	V	p	Yes
T	p	V	?

Not in Atkins

From $p(V)$ to $V(p)$



(Image: http://casey.brown.edu/chemistry/misspelled-research/crp/Edu/Documents/00_Chem201/7_phase_equil_1/Chem201_7_phase_equil_1.htm)

Not in Atkins

Gibbs Energy of vdW Fluid

$$V = \left(\frac{\partial G}{\partial p} \right)_{T,n}$$

At constant T and n ,

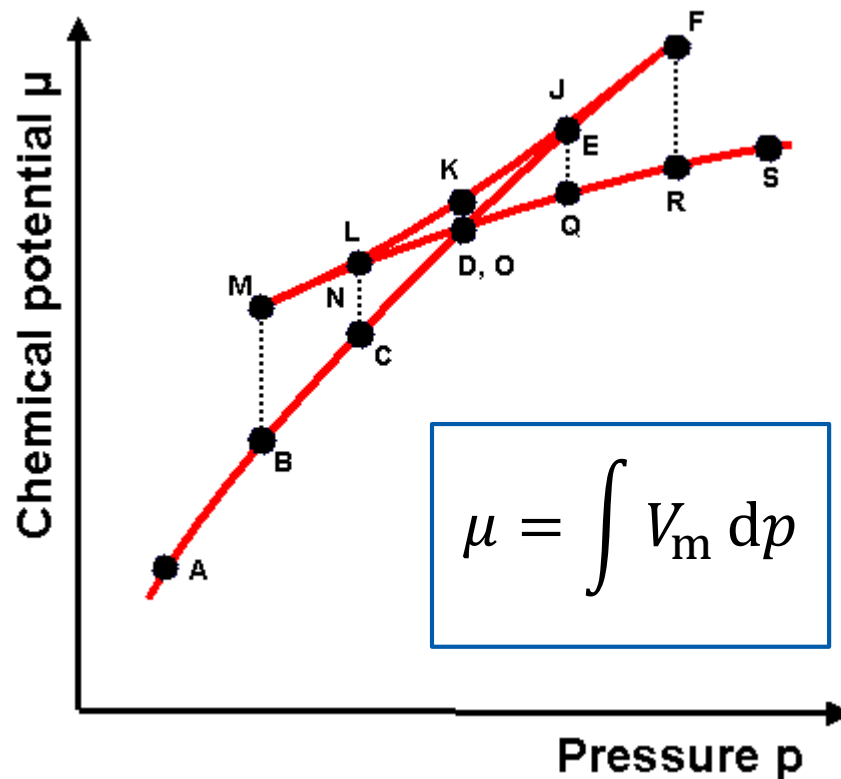
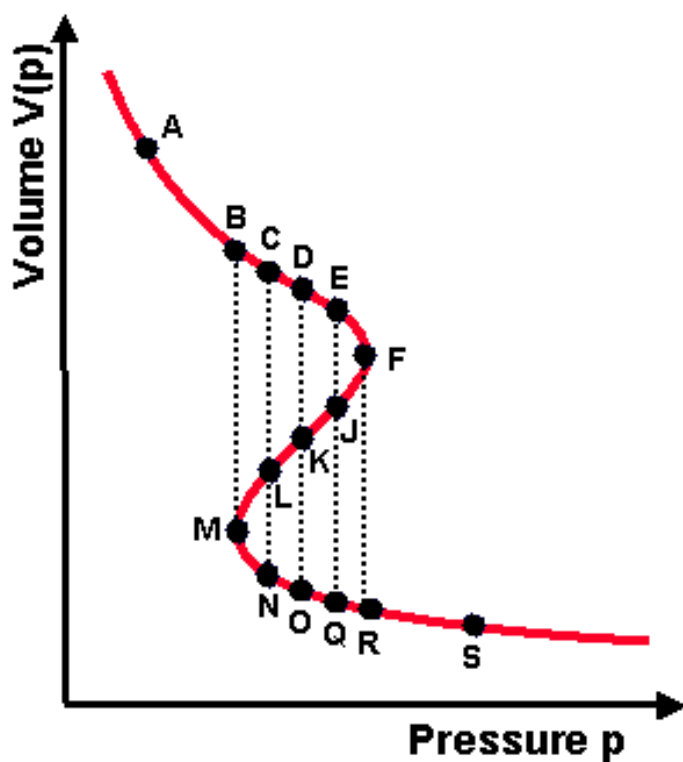
$$G = \int V \, dp$$

or,

$$\mu = \int V_m \, dp$$

Not in Atkins

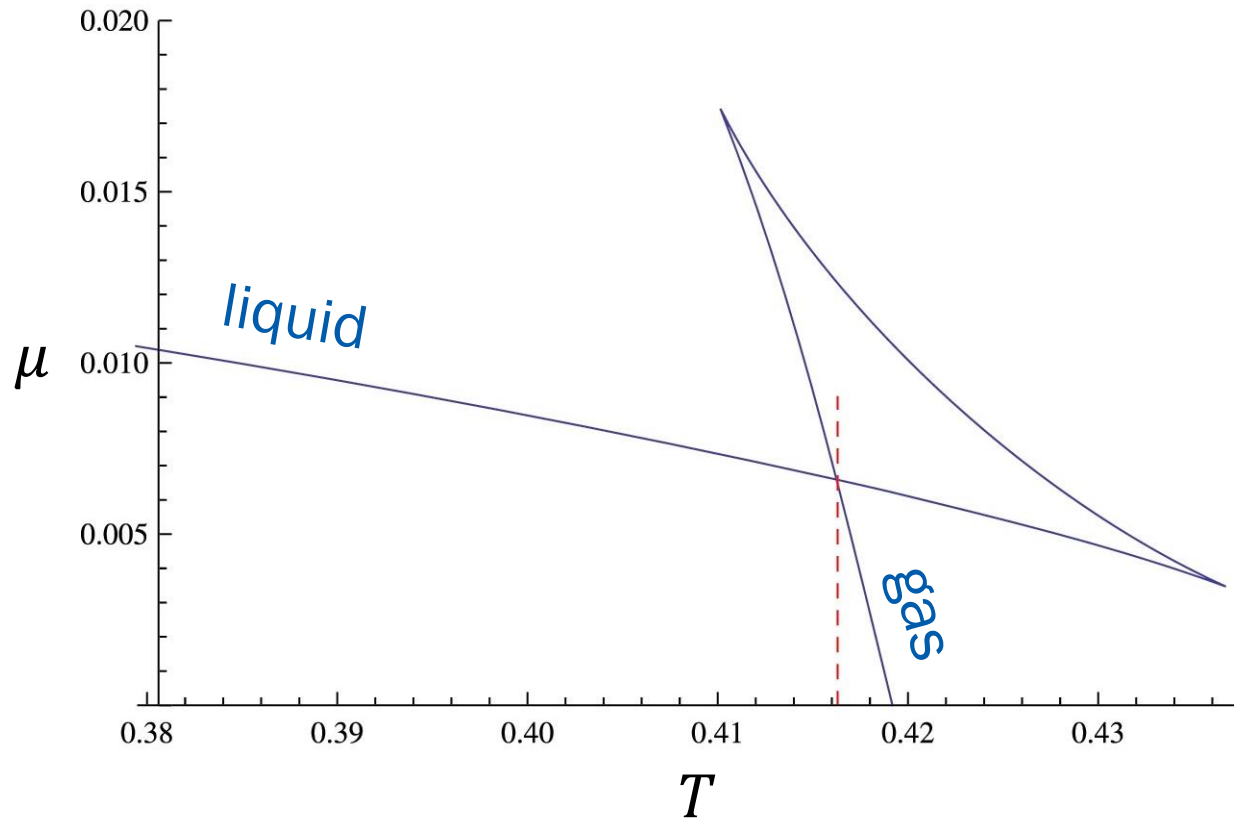
Chemical Potential $\mu(p)$



$$\mu = \int V_m dp$$

Not in Atkins

$\mu(T)$ of vdW Fluid



(Image: He et al., *Nuclear Physics B* 915: 243-261 (2017), Figure 3)

Generalization

Use **thermodynamics** of chemical potentials
to study general properties of phase transitions

1. Phase transitions on (T, p) -space
2. Location of phase boundaries

Use of Gibbs-Duhem Equation

Recall

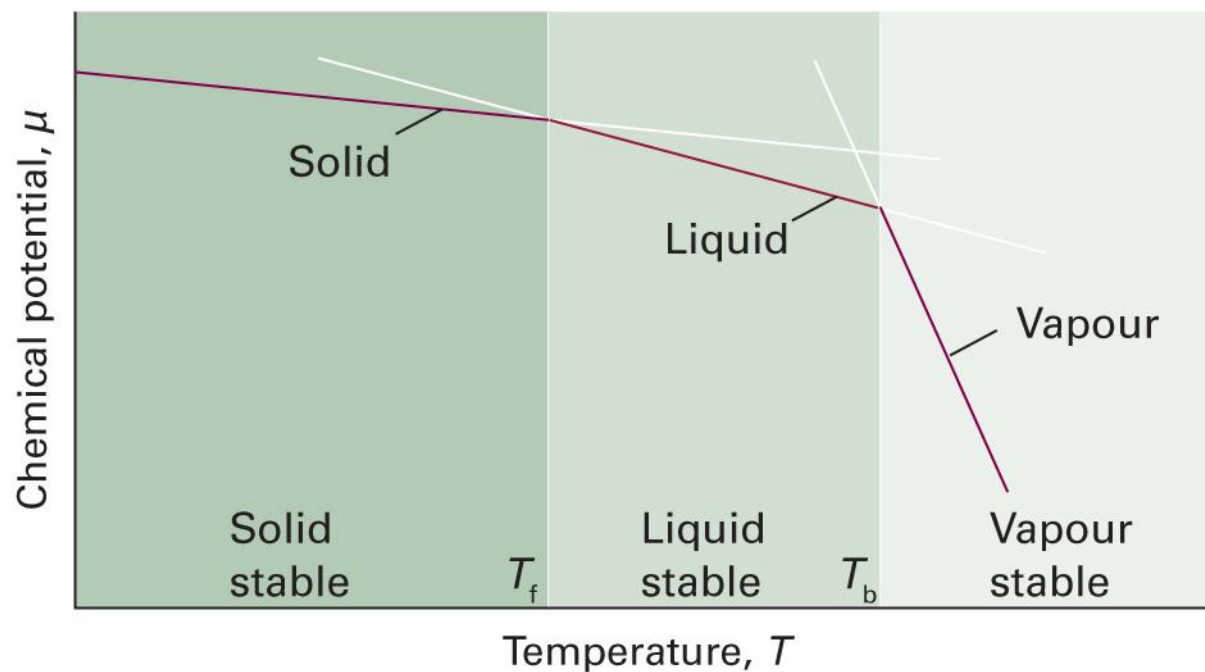
$$d\mu = -S_m dT + V_m dp$$

which gives

$$\left(\frac{\partial \mu}{\partial T}\right)_{p,n} = -S_m, \quad \left(\frac{\partial \mu}{\partial p}\right)_{T,n} = V_m$$

Chemical Potential and T

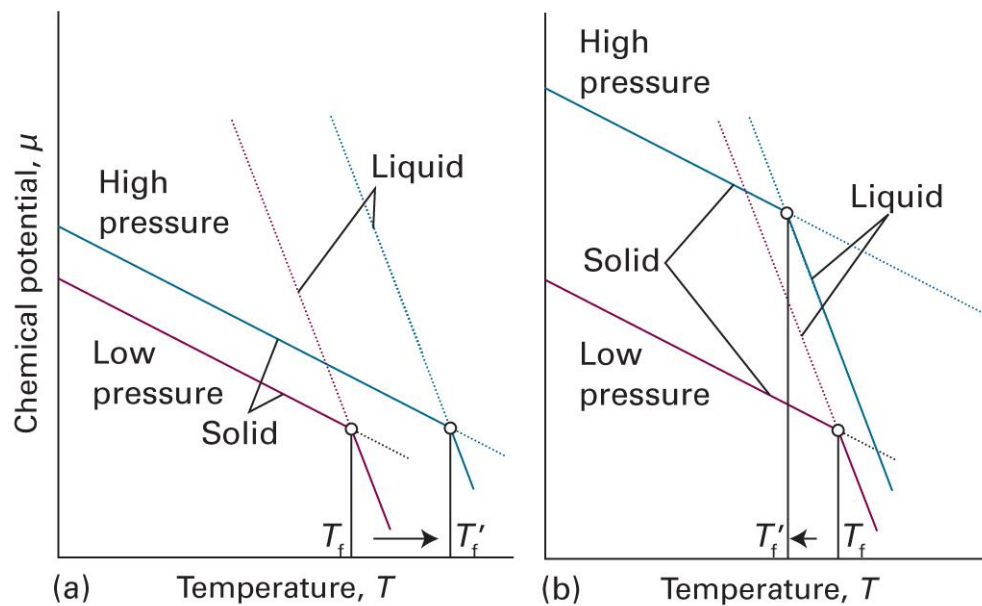
$$\left(\frac{\partial \mu}{\partial T}\right)_{p,n} = -S_m$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.1)

Chemical Potential and p

$$\left(\frac{\partial \mu}{\partial p}\right)_{T,n} = V_m$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.2)

Phase Boundaries

When two phases α and β are at equilibrium,

$$\mu(\alpha; p, T) = \mu(\beta; p, T)$$

For infinitesimal changes dp and dT ,

$$d\mu(\alpha) = d\mu(\beta)$$

Since $d\mu = V_m dp - S_m dT$ (Gibbs-Duhem equation),

$$V_m(\alpha)dp - S_m(\alpha)dT = V_m(\beta)dp - S_m(\beta)dT$$

Clapeyron Equation

$$V_m(\alpha)dp - S_m(\alpha)dT = V_m(\beta)dp - S_m(\beta)dT$$

Rearrangement leads to

$$\{S_m(\beta) - S_m(\alpha)\}dT = \{V_m(\beta) - V_m(\alpha)\}dp$$

$$\Delta_{\text{trs}}SdT = \Delta_{\text{trs}}Vdp$$

Thus,

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}S}{\Delta_{\text{trs}}V}, \quad \frac{dT}{dp} = \frac{\Delta_{\text{trs}}V}{\Delta_{\text{trs}}S}$$

(Almost Always) Positive Slope

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}S}{\Delta_{\text{trs}}V}$$

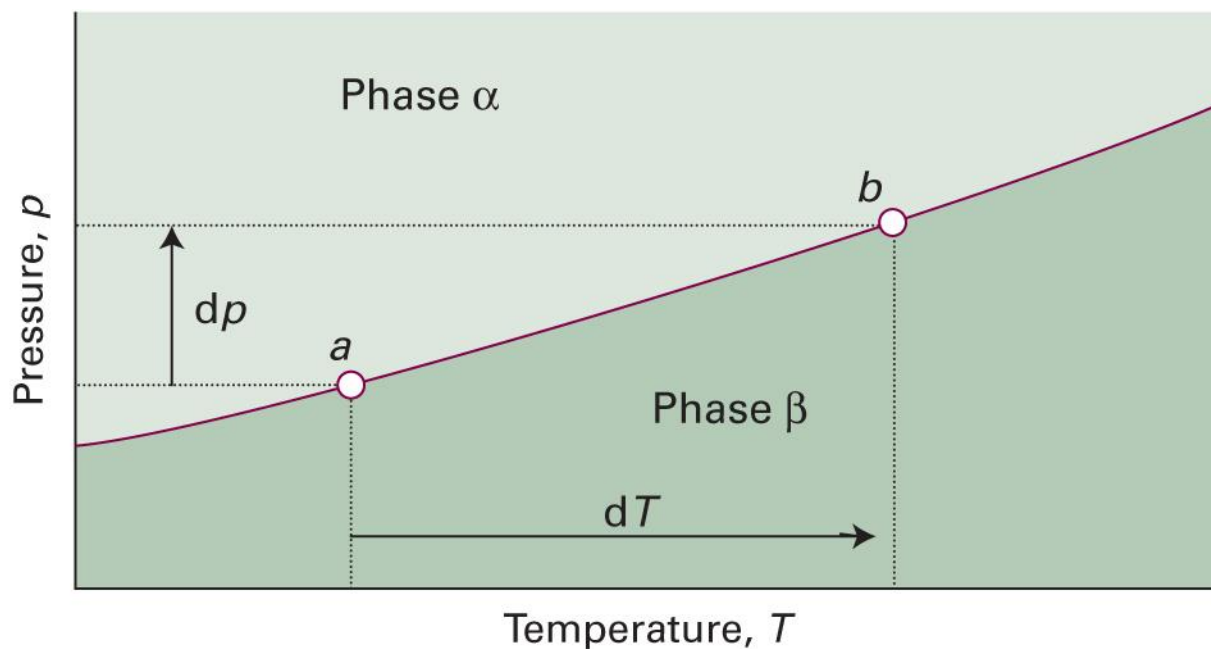
For a phase transition, $\Delta_{\text{trs}}S = \Delta_{\text{trs}}H/T$, and

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}H}{T\Delta_{\text{trs}}V}$$

Hence, dp/dT is almost always positive (unless $\Delta_{\text{trs}}V < 0$)

Phase Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}}H}{T\Delta_{\text{trs}}V}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.4)

Solid-Liquid Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{fus}}H}{T\Delta_{\text{fus}}V}$$

Assuming $\Delta_{\text{fus}}H$ and $\Delta_{\text{fus}}V$ are constant,

$$dp = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \frac{1}{T} dT$$

$$\int_{p_1}^{p_2} dp = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \int_{T_1}^{T_2} \frac{1}{T} dT$$

Solid-Liquid Boundary

$$\int_{p_1}^{p_2} dp = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \int_{T_1}^{T_2} \frac{1}{T} dT$$

After integration,

$$p_2 - p_1 = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \ln \left(\frac{T_2}{T_1} \right)$$

Solid-Liquid Boundary

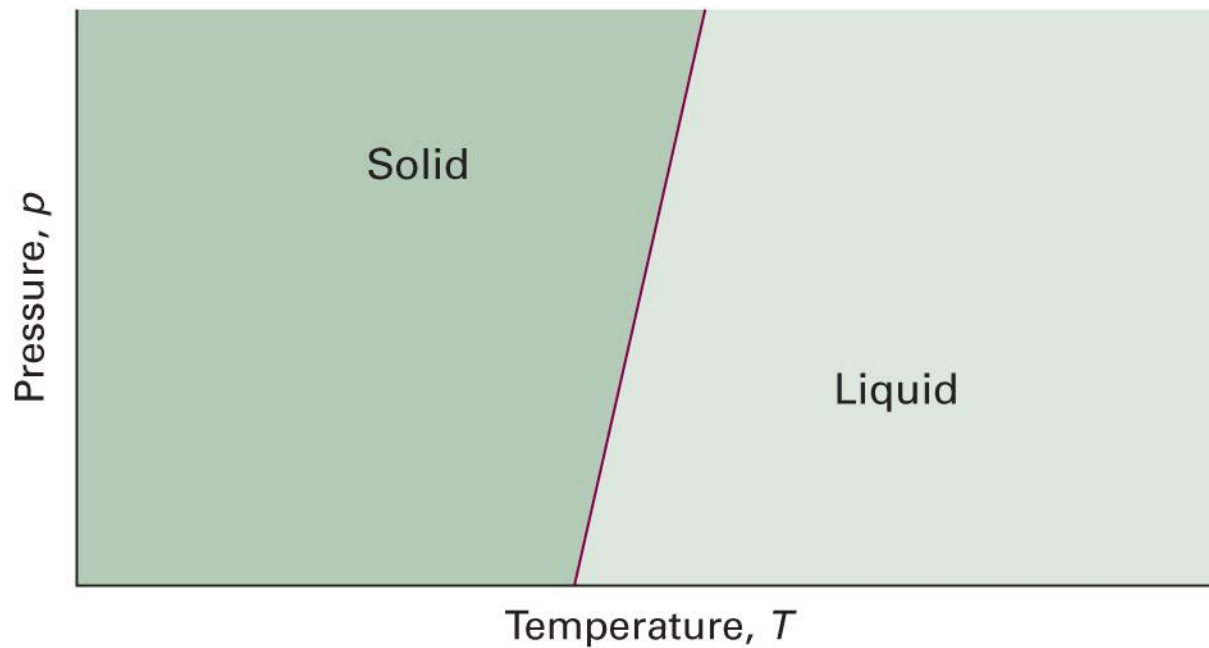
$$p_2 - p_1 = \frac{\Delta_{\text{fus}}H}{\Delta_{\text{fus}}V} \ln \left(\frac{T_2}{T_1} \right)$$

If $T_1 \approx T_2$, using $\ln(1 + x) = x - x^2/2 + \dots$,

$$\ln \left(\frac{T_2}{T_1} \right) = \ln \left(1 + \frac{T_2 - T_1}{T_1} \right) \approx \frac{T_2 - T_1}{T_1}$$

$$p_2 - p_1 = \frac{\Delta_{\text{fus}}H}{T_1 \Delta_{\text{fus}}V} (T_2 - T_1)$$

Solid-Liquid Boundary



Liquid-Gas Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{vap}}H}{T\Delta_{\text{vap}}V}$$

Since $V_{\text{m}}(g) \gg V_{\text{m}}(l)$,

$$\Delta_{\text{vap}}V = V_{\text{m}}(g) - V_{\text{m}}(l) \approx V_{\text{m}}(g) \approx RT/p$$

Thus,

$$\frac{1}{p} \frac{dp}{dT} = \frac{\Delta_{\text{vap}}H}{RT^2}$$

(Clausius-Clapeyron equation)

Liquid-Gas Boundary

$$\frac{1}{p} dp = \frac{\Delta_{\text{vap}}H}{RT^2} dT$$

Assuming $\Delta_{\text{vap}}H$ is constant,

$$\int_{p_1}^{p_2} \frac{1}{p} dp = \frac{\Delta_{\text{vap}}H}{R} \int_{T_1}^{T_2} \frac{1}{T^2} dT$$

$$\ln \left(\frac{p_2}{p_1} \right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Liquid-Gas Boundary

$$\ln \left(\frac{p_2}{p_1} \right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

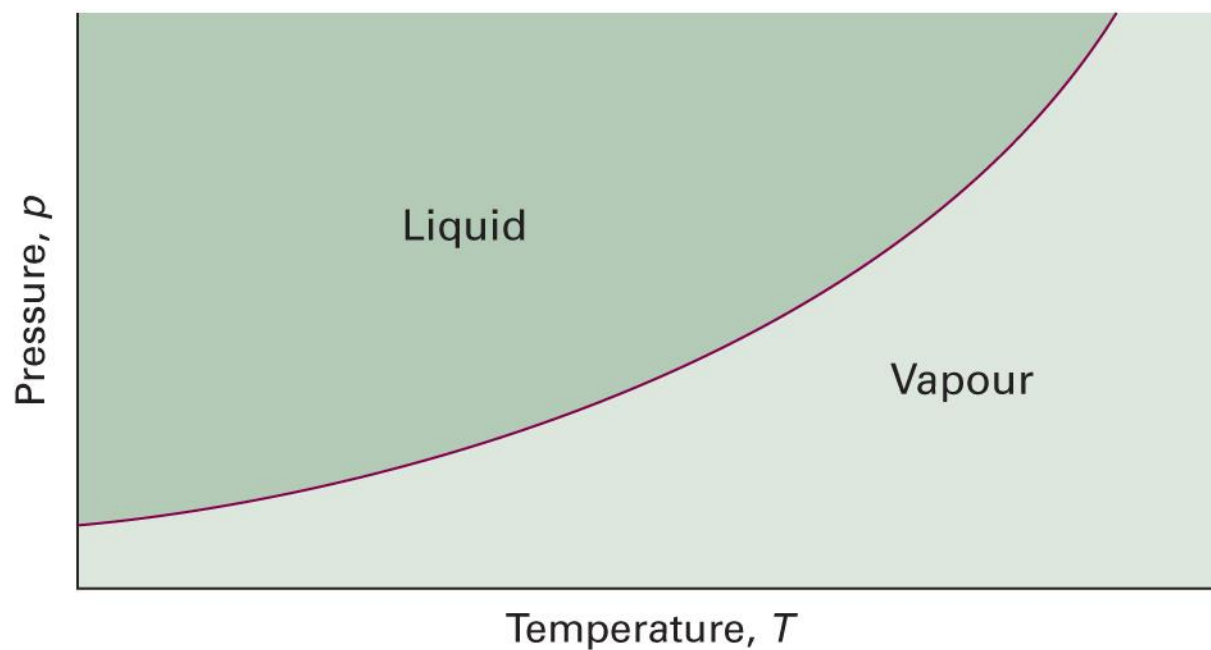
Hence,

$$p_2 = p_1 e^{A-B/T_2}$$

where

$$A = \frac{\Delta_{\text{vap}}H}{RT_1}, \quad B = \frac{\Delta_{\text{vap}}H}{R}$$

Liquid-Gas Boundary



Solid-Gas Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{sub}}H}{T\Delta_{\text{sub}}V}$$

Similarly, $\Delta_{\text{sub}}V \approx V_{\text{m}}(g) \approx RT/p$, and

$$\frac{1}{p} \frac{dp}{dT} = \frac{\Delta_{\text{sub}}H}{RT^2}$$

Assuming $\Delta_{\text{sub}}H$ is constant,

$$\ln \left(\frac{p_2}{p_1} \right) = \frac{\Delta_{\text{sub}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

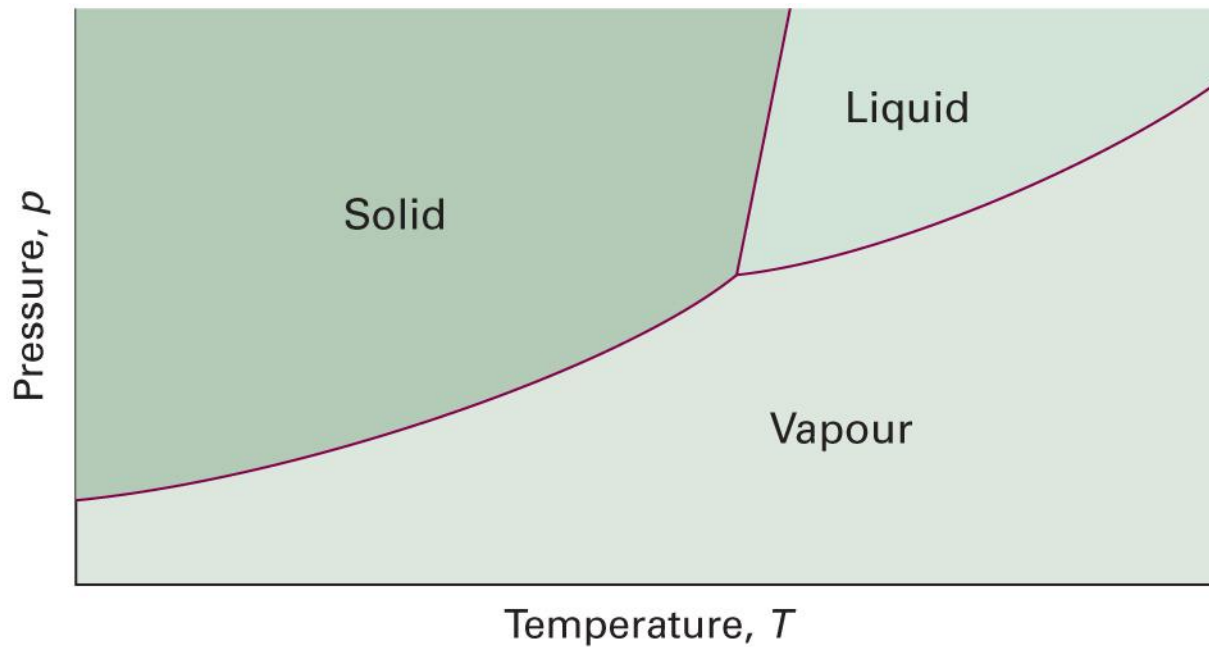
Solid-Gas vs. Liquid-Gas

$$\text{Solid-gas: } \ln\left(\frac{p_2}{p_1}\right) = \frac{\Delta_{\text{sub}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\text{Liquid-gas: } \ln\left(\frac{p_2}{p_1}\right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

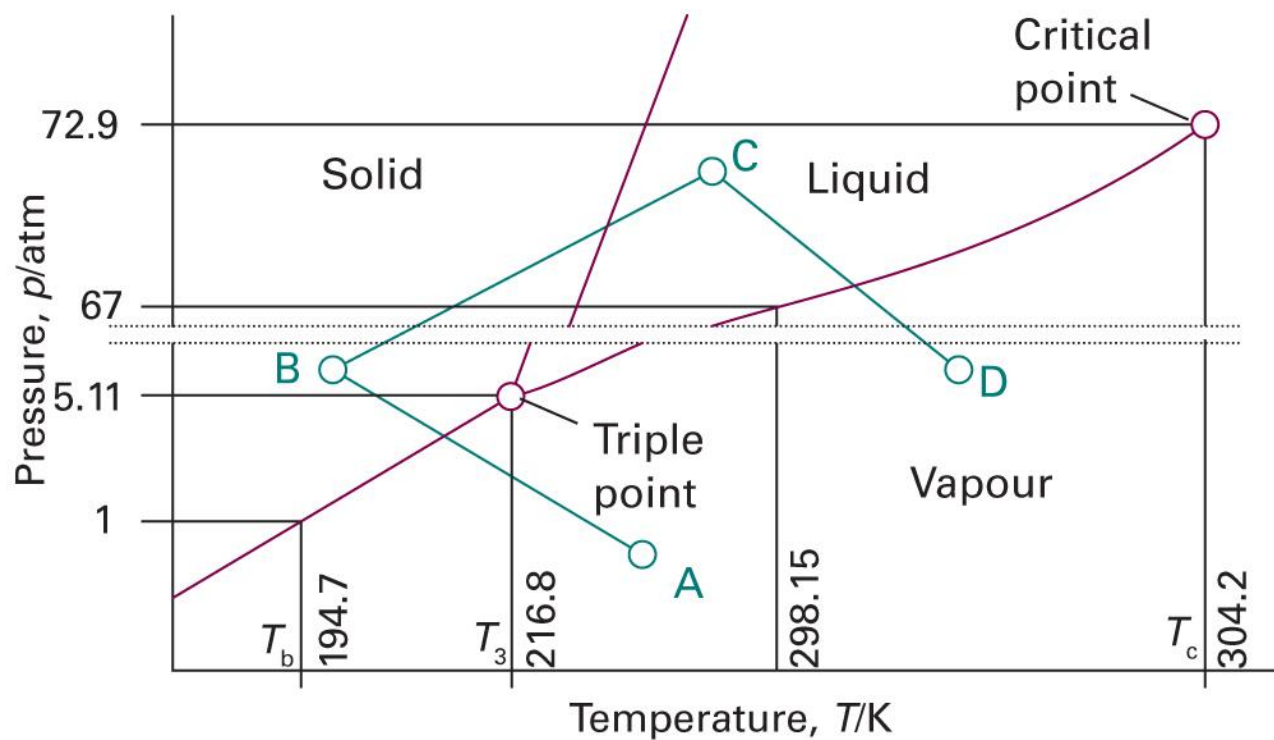
Since $\Delta_{\text{sub}}H = \Delta_{\text{fus}}H + \Delta_{\text{vap}}H > \Delta_{\text{vap}}H$, the sublimation curve has a steeper slope than the vaporization curve

Phase Diagram



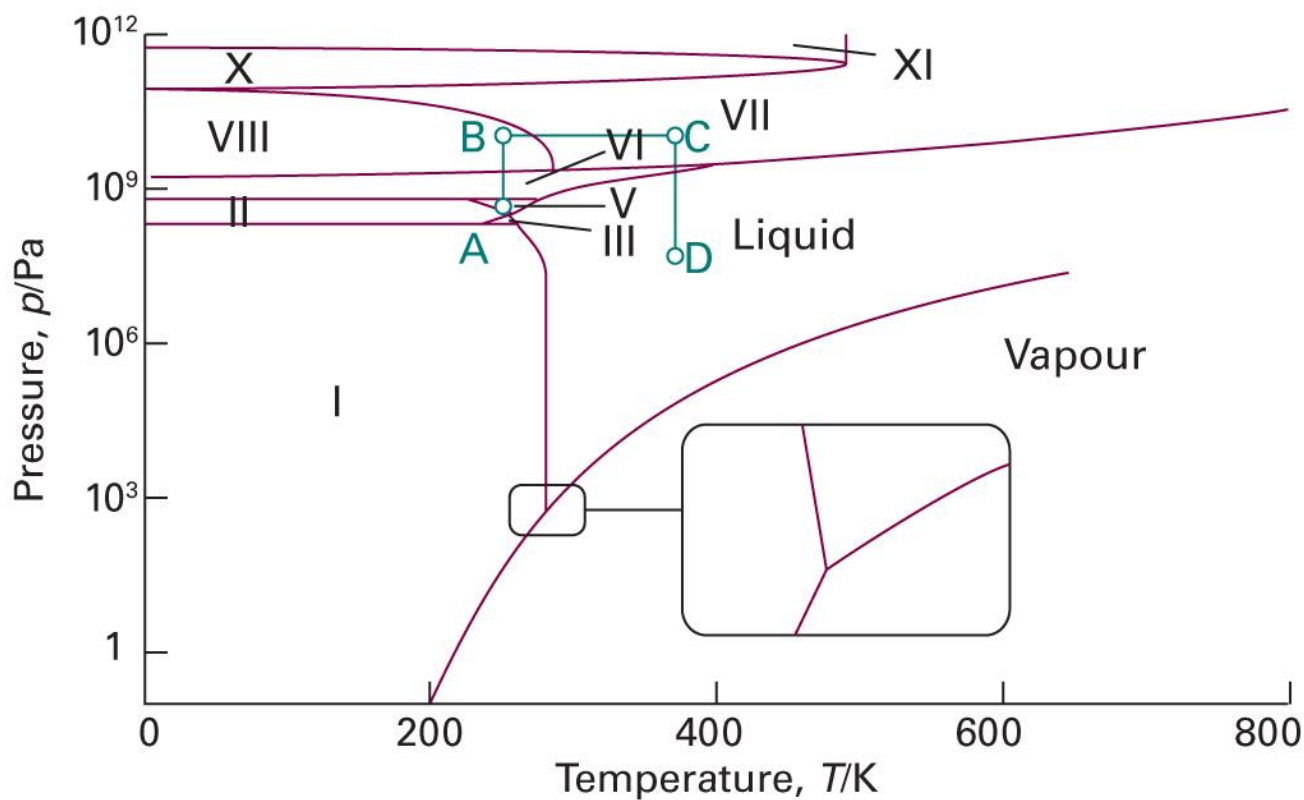
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.7)

Revisiting CO₂



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.8)

Revisiting H₂O



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.9)

Summary

